Simple and compound gear trains

National 5 · Mechanisms & structures · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Calculate the velocity ratio.
- 45 ÷ 15 = 3

1b) Calculate the speed of B.
- 900 ÷ 3 = 300 revs min^{-1}

2a) Calculate the velocity ratio.
- 60 ÷ 20 = 3

2b) Calculate the output speed.
- 720 ÷ 3 = 240 revs min^{-1}

3a) Calculate the velocity ratio.
- (75 ÷ 25) × (80 ÷ 20) = 3 × 4 = 12

3b) Calculate the output speed of D.
- 1000 ÷ 12 = 83 revs min^{-1}

4a) Calculate the velocity ratio.
- 40 ÷ 10 = 4

4b) Calculate the output speed.
- 600 ÷ 4 = 150 revs min^{-1}

4c) State the direction of the output compared with the driver.
- Opposite direction

5a) Calculate the velocity ratio.
- (64 ÷ 16) × (72 ÷ 24) = 4 × 3 = 12

5b) Calculate the output shaft speed.
- 1500 ÷ 12 = 125 revs min^{-1}

6a) Calculate the velocity ratio.
- B is the driven gear: VR = 10 ÷ 30 = 0.33 (1 : 3)

6b) Calculate the speed of B.
- 300 × 3 = 900 revs min^{-1}

7a) Calculate the velocity ratio.
- (72 ÷ 18) × (60 ÷ 24) = 4 × 2.5 = 10

7b) Calculate the output speed.
- 1200 ÷ 10 = 120 revs min^{-1}

8a) Calculate the velocity ratio.
- 36 ÷ 12 = 3

8b) Calculate the speed of the driven gear.
- 1800 ÷ 3 = 600 revs min^{-1}

8c) State one reason for using plastic gears in the model.
- Cheaper, lighter or self-lubricating (less friction and heat)

9a) Calculate the overall velocity ratio.

- (70 ÷ 14) × (80 ÷ 20) = 5 × 4 = 20

9b) Calculate the motor speed needed.

- 20 × 20 = 400 revs min^{-1}